

Investigating Weak-to-Strong Generalization through In-Context Learning on the MATH Dataset

Samuel Lee
hsl023@ucsd.edu

Zhen Wang
zhw085@ucsd.edu

Abstract

Weak-to-strong generalization (W2S) aims to supervise advanced language models using weaker ones. This offers a scalable alternative to human oversight. This work investigates W2S through in-context learning on the MATH dataset by prompting strong models with examples created by weaker models. We use the Performance Gap Recovered (PGR) metric to evaluate several OpenAI models. Our findings show that strong models often perform similarly to their best results under weak supervision, with PGR values usually close to 1. However, increasing the number of weak examples does not always improve performance. In fact, noisy examples can make results less stable. These outcomes indicate that weak supervision can be effective, but it heavily relies on the structure and quality of the examples given.

Code: <https://github.com/samuelllee77/weak2strong>

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1 Introduction

Recent advances in LLMs show a promising future for AI. For example, the study by [Jones and Bergen \(2025\)](#) found that GPT-4.5 has passed the Turing test. This indicates that it behaves nearly as intelligently, or indistinguishably, as a human. As AI systems keep advancing, future models are likely to surpass human intelligence in many areas. This creates a key challenge: how can humans effectively supervise or guide AI systems that are smarter than us? To address this question, we need to develop scalable forms of supervision that do not require perfect human oversight. One promising method is weak-to-strong generalization (W2S). This involves training or guiding stronger models using feedback or examples from weaker models. The main idea is that a strong model can learn from imperfect supervision, gaining valuable insights even if its teacher does not perform well, as its understanding of this concept has direct implications for AI safety, interpretability, and scalability.

This project looks at weak-to-strong generalization by focusing on in-context learning, which refers to the ability of large language models (LLMs) to change their behavior based on examples in their prompts. To measure how well a strong model can regain performance from weak supervision, we use the Performance Gap Recovered (PGR) metric, which shows how much of the strong model’s performance with gold labels can be restored when using weakly generated few-shot examples.

2 Literature Review

Few shot Prompts In the paper introducing GPT-3, [Brown et al. \(2020\)](#) demonstrated that large language models are effective few-shot learners. By including a few solved examples in the prompt, a model’s accuracy can increase substantially. This finding establishes in-context learning as a mechanism through which language models can simulate supervised learning within a single forward pass, without any gradient updates. However, subsequent research by [Reynolds and McDonell \(2021\)](#) showed that, in some cases, well-designed zero-shot prompts can match or even surpass the performance of few-shot prompts. They argued that the role of examples is not to teach the model during inference, but rather to help it locate an already learned task within its internal representation space.

Weak to Strong Generalization Weak-to-strong generalization was first proposed as a method for aligning advanced models with minimal human supervision, as discussed in OpenAI’s report on Weak-to-Strong Generalization ([Burns et al. 2023](#)). The framework posits that if a stronger model can successfully interpret and learn from the outputs of a weaker model, it may eventually generalize beyond its teacher, mirroring how humans learn from imperfect examples. More recently, [Lang, Sontag and Vijayaraghavan \(2024\)](#) provided the theoretical foundation for this phenomenon. Their paper formalizes the intuition that, as long as weak supervision contains partial truth and strong models are expressive enough to identify it, weak-to-strong generalization is possible and theoretically predictable.

3 Methods

3.1 Data

All experiments use the MATH dataset, which is designed to evaluate mathematical reasoning in areas like algebra, geometry, and number theory. Each problem comes with a step-by-step solution and a final numeric or symbolic answer. This setup allows for accurate grading through automated answer normalization. The dataset is split into a training set for creating few-shot prompts and a separate test set for evaluation. The evaluation pipeline uses the official `grade_answer()` function from the PRM800K framework, which ensures consistent comparisons across models. This dataset provides an ideal environment for studying weak-to-strong generalization. It requires structured reasoning instead of just pattern recognition, allows for detailed accuracy measurement, and shows how well a strong model can learn from weakly generated demonstrations.

3.2 Determining Strong and Weak Models

We evaluate using a set of OpenAI models:

- gpt-5-mini
- gpt-4.1-mini
- o3-mini
- gpt-4.1-nano
- gpt-4o-mini

We first determine the relative strength of each model by evaluating individual accuracies on the MATH subset using a standardized gold prompt. For a model M , its baseline accuracy is denoted

$$A_M^{\text{gold}}.$$

A model M_s is designated as a *strong* model for M_w if and only if

$$A_{M_s}^{\text{gold}} > A_{M_w}^{\text{gold}}.$$

Only such weak→strong pairings are considered in subsequent cross-model experiments.

3.3 Cross-Model Weak-to-Strong Experiments

For each valid weak-to-strong model pair (M_w, M_s) , where the strong model attains strictly higher gold-prompt accuracy than the weak model, we conduct in-context learning experiments to assess whether the strong model can benefit from weak demonstrations.

Given a problem input x , the weak model produces a predicted solution.

$$\hat{y}_w = M_w(x).$$

A demonstration set of size k is then constructed by sampling k input–output pairs generated by the weak model:

$$D_w^{(k)} = \{(x_1, \hat{y}_{w,1}), \dots, (x_k, \hat{y}_{w,k})\}.$$

The strong model is prompted with these demonstrations and queried on the target input x , yielding.

$$\hat{y}_s^{(k)} = M_s(x \mid D_w^{(k)}).$$

In addition, we evaluate the impact of the number of few-shot examples on PGR, corresponding to $k = 0$, $k = 2$, and $k = 4$, to measure how the number of weak demonstrations influences weak-to-strong generalization.

3.4 Performance Metric

We quantify weak-to-strong generalization using the **Performance Gap Recovered (PGR)** metric, which measures the extent to which weak supervision helps the strong model close the performance gap relative to gold supervision. PGR is defined as:

$$\text{PGR} = \frac{A_{S|\text{weak}} - A_{S|\text{gold}}}{A_{W|\text{gold}} - A_{S|\text{gold}}}.$$

Here,

- $A_{S|\text{weak}}$ is the accuracy of the strong model when prompted with weak-labeled demonstrations,
- $A_{S|\text{gold}}$ is the accuracy of the strong model when prompted with gold (correct) demonstrations,
- $A_{W|\text{gold}}$ is the accuracy of the weak model when prompted with gold demonstrations.

A value of 1 indicates that the strong model fully recovers the weak model’s performance advantage. Values greater than 1 indicate that weak supervision improves the strong model beyond the weak model’s own performance, whereas values below 1 indicate only partial recovery.

4 Results

All experiments were conducted over 5 independent trials, each using a different random sample of 200 MATH test problems. This repetition reduces variance due to dataset sampling and makes cross-model comparisons more reliable.

4.1 Model Performance on MATH

We first evaluated each model on 200 MATH problems using gold prompts. This helped us rank performance and identify valid weak and strong pairs. As shown in Figure 2 and

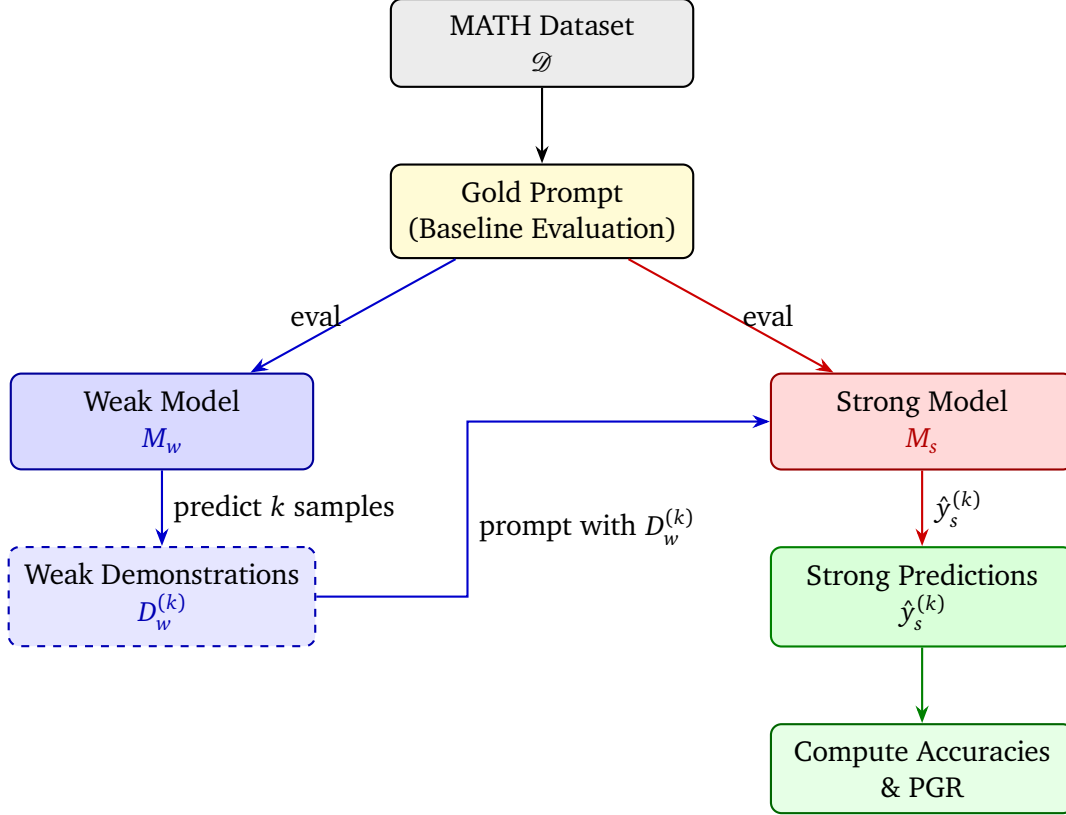


Figure 1: Weak-to-Strong (W2S) evaluation pipeline.

Table 1, accuracy varied significantly across models. GPT-5-mini achieved the highest average performance, closely followed by gpt-4.1-mini and o3-mini. The two smallest models, gpt-4o-mini and gpt-4.1-nano, performed much worse.

Table 1: Baseline model accuracy on the MATH dataset (mean $\pm$ standard deviation).

Model	Mean Acc $\pm$ Std
gpt-5-mini	0.947 ± 0.015
o3-mini	0.837 ± 0.012
gpt-4.1-mini	0.917 ± 0.021
gpt-4o-mini	0.523 ± 0.006
gpt-4.1-nano	0.780 ± 0.017

4.2 Cross-Model Weak-to-Strong Experiments

We ran all weak and strong combinations where the strong model had higher gold-prompt accuracy. The average PGR values from these pairings are shown in Figure 3 and Table 2.

In almost all valid combinations, PGR stayed close to 1, and all combinations have PGR

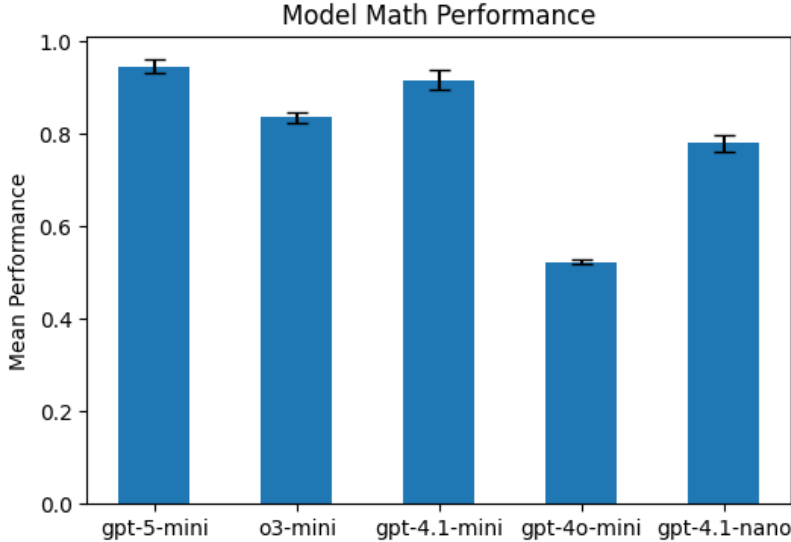


Figure 2: Baseline model accuracy on the MATH dataset. This establishes the weak/strong hierarchy used in cross-model experiments.

larger than 0.8, except gpt-4.1-nano and o3-mini, for which PGR is 0.75. This shows that strong models often matched their own gold-prompt accuracy, even when prompted with only weak demonstrations.

Table 2: Mean PGR ($\pm$ standard deviation) for each weak $\rightarrow$ strong model pair.

Strong Model	gpt-4.1-mini	o3-mini	gpt-4.1-nano	gpt-4o-mini
gpt-5-mini	1.000 ± 0.000	0.889 ± 0.157	0.926 ± 0.105	0.967 ± 0.047
gpt-4.1-mini	N/A	0.867 ± 0.189	1.000 ± 0.050	0.965 ± 0.050
o3-mini	N/A	N/A	0.756 ± 0.175	0.872 ± 0.181
gpt-4.1-nano	N/A	N/A	N/A	0.974 ± 0.036

4.3 Few-Shot Experiments ($k = 0, 2, 4$)

We next examined how the number of weak demonstrations affects W2S performance. In this experiment, we chose gpt-4.1-mini as the weak model and gpt-5-mini as the strong model. The few-shot PGR results, shown in Figure 4 and Table 3, reveal no consistent monotonic trend. Surprisingly, 0-shot prompts achieved the highest mean PGR, matching or outperforming 2-shot and 4-shot prompts. The 2-shot condition exhibited the greatest variance and lower mean PGR, suggesting that introducing weak examples may also introduce noise or misleading reasoning chains that destabilize performance. The 4-shot setting partially recovered performance, but did not surpass the 0-shot baseline.

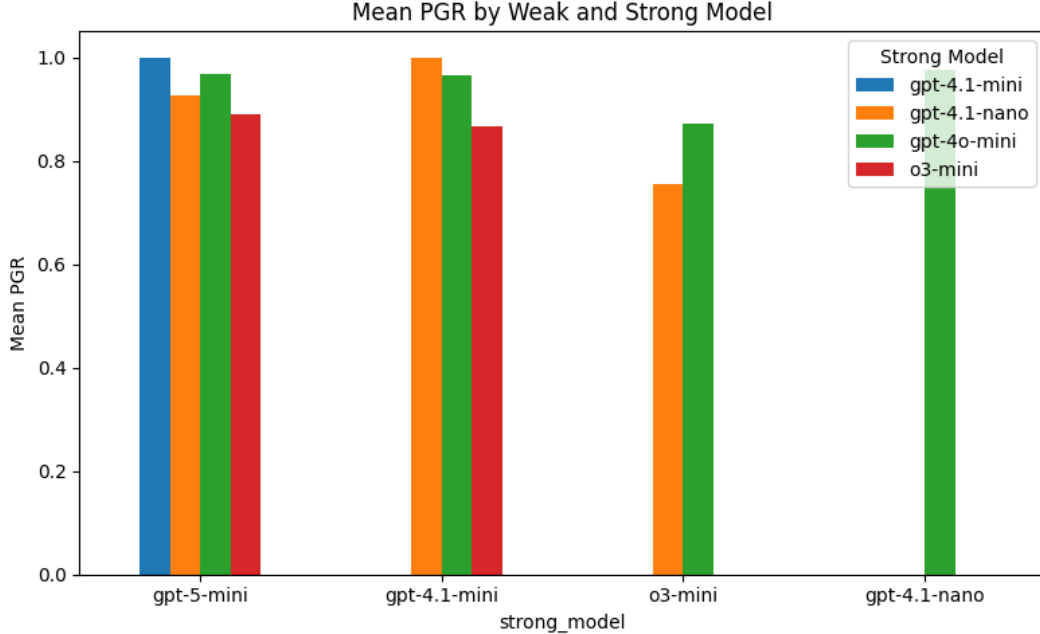


Figure 3: Mean PGR across all valid weak-to-strong model pairs.

Table 3: Few-shot Performance Gap Recovered (PGR) for different numbers of weak demonstrations (k).

Shots (k)	Mean Acc $\pm$ Std
0	1.000 ± 0.000
2	0.831 ± 0.150
4	0.933 ± 0.115

5 Discussion

Our experiments show that strong language models can effectively use weak demonstrations through in-context learning. They often achieve PGR values close to 1 in the 5-trial evaluations. This indicates that weak supervision can bring back most of the strong model’s performance advantage compared to gold-labeled examples. In some cases, PGR goes above 1. This supports the idea that demonstrations serve as task-guiding cues instead of just sources of correct reasoning.

However, the extended 5-trial analysis also points out the instability caused by weak examples. The 2-shot condition, in particular, shows large variance across repeated runs. This suggests that weak demonstrations may introduce inconsistent reasoning patterns that the strong model mimics. Since LLMs tend to base their reasoning on the structural patterns in few-shot prompts, even partially incorrect or noisy weak demonstrations can skew the model’s problem-solving path. This explains why 0-shot prompts, which lack such anchors, sometimes perform better than multi-shot prompts.

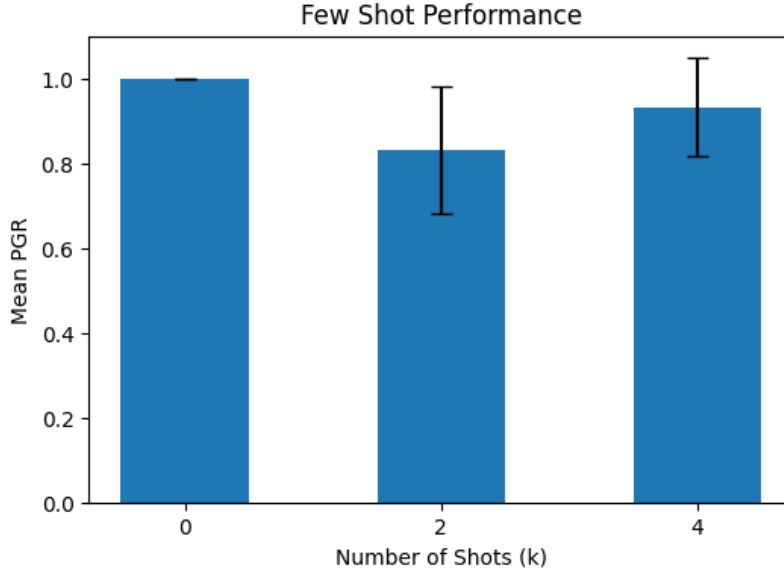


Figure 4: Comparison of few-shot performance (0, 2, 4 shots).

Interestingly, the similarity in PGR among strong models implies that W2S behavior relies more on how models handle contextual examples than on their baseline ability. Even models with significant differences in accuracy acted similarly in terms of how much performance they could regain under weak supervision.

6 Conclusion

Overall, our evaluation of five trials shows that weak-to-strong generalization through in-context learning is both promising and fragile. Strong models often regain most of their gold-prompt performance when guided by weak demonstrations, with PGR values frequently close to 1. This suggests that even imperfect supervision can offer useful task hints.

However, these findings point out several limitations of our current study. Our experiments used relatively small evaluation sets per trial, relied on unfiltered weak demonstrations, and did not control for differences in chain-of-thought or decoding settings. Additionally, the analysis focused on a single dataset (MATH), which may not apply to tasks that involve natural language, planning, or multi-turn reasoning. These limitations mean that although our results show important patterns, they should be taken with caution.

Future work should aim to reduce demonstration noise by filtering weak samples using correctness or confidence guidelines and exploring more structured prompting formats. Expanding to multiple datasets and larger evaluation sets will also help clarify when weak-to-strong generalization works and when it doesn't. Improving these areas is crucial for creating more stable and scalable methods of weak supervision in AI systems.

References

- Brown, Tom, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell et al.** 2020. “Language models are few-shot learners.” *Advances in neural information processing systems* 33: 1877–1901
- Burns, Collin, Pavel Izmailov, Jan Hendrik Kirchner, Bowen Baker, Leo Gao, Leopold Aschenbrenner, Yining Chen, Adrien Ecoffet, Manas Joglekar, Jan Leike et al.** 2023. “Weak-to-strong generalization: Eliciting strong capabilities with weak supervision.” *arXiv preprint arXiv:2312.09390*
- Jones, Cameron R, and Benjamin K Bergen.** 2025. “Large language models pass the turing test.” *arXiv preprint arXiv:2503.23674*
- Lang, Hunter, David Sontag, and Aravindan Vijayaraghavan.** 2024. “Theoretical analysis of weak-to-strong generalization.” *Advances in neural information processing systems* 37: 46837–46880
- Reynolds, Laria, and Kyle McDonell.** 2021. “Prompt programming for large language models: Beyond the few-shot paradigm.” In *Extended abstracts of the 2021 CHI conference on human factors in computing systems*.